

12.2 Vectors

1 Warm Up

Question: How would you describe to someone, how to walk from the point $(2, 4)$ to $(-3, -3)$?

2 Motivation

When formalizing mathematically the idea of movement and change, we need more than simply numbers. Vectors allow us to represent movement in a certain direction for a certain distance or magnitude. Lots of interesting mathematical properties unfold from the simple definitions regarding vectors. We first need lots of definitions to unpack.

3 Definitions & Theorems

1. A **vector** has both a direction and length (magnitude) we can think of it starting at an **initial point** and ending at a **terminal point**. If vector u and v have the same length and direction we say $u = v$.
2. **Adding** vectors $u + v$ can be thought of as first traveling down u and then traveling down v . We can imagine moving the initial point of v to the terminal point of u (Tip to Tail). Does it matter $u + v$ vs. $v + u$ (Parallelogram Law)?
3. Since $u + u =$ "two" u 's we think of 2 as a **scalar**. Hence we can multiply vectors by scalars cv . Note that two vectors are "parallel" if and only if they are scalar multiples of each other.
4. The **negative** of a vector is like "reversing" the vector. Thus, $u - v = u + (-v)$.
5. **Component form** of a vector has its initial point at the origin and looks like $\langle x, y \rangle$. In three dimensions it looks like $\langle x, y, z \rangle$. These vectors are called **position vectors**. Don't confuse vectors with points (although they are related).
6. Given two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$, the **vector representation** from A to B is $\langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$.
7. The length or **magnitude** of a vector $v = \langle x, y \rangle$ is $|v| = \sqrt{x^2 + y^2}$. In three dimensions if $v = \langle x, y, z \rangle$, $|v| = \sqrt{x^2 + y^2 + z^2}$.
8. $\langle x_1, y_1, z_1 \rangle + \langle x_2, y_2, z_2 \rangle = \langle x_1 + x_2, y_1 + y_2, z_1 + z_2 \rangle$ and $c\langle x, y, z \rangle = \langle cx, cy, cz \rangle$.

9. $\langle 1, 0, 0 \rangle, \langle 0, 1, 0 \rangle, \langle 0, 0, 1 \rangle$ are called **standard basis vectors** and are denoted i, j, k . Every vector can be represented as the summation of scalar multiples of these basis vectors.
10. A **unit vector** has magnitude 1. If a is some vector, $a/|a|$ is the unit vector of a .

4 Examples

Problem 1. *Add the vectors u and v . Find $3u$. Find $u - 2v$.*

Problem 2. *Find the vector representation AB given the points $A(1, -5, 3)$ and $B(6/5, 4, -2)$.*

Problem 3. *If $u = \langle 2, -3, -4 \rangle$ and $v = \langle 2, 4, -1 \rangle$ find $|2u - 3v|$.*

Problem 4. *Find the standard basis representation of $\langle -7.2, 3.4, -10.1 \rangle$.*

Problem 5. *Find the unit vector in the same direction as the vector $4i - 3j - k$.*

5 Scratch Work